

SELF-FOCUSING OF FRÖHLICH WAVES AND CYTOSKELETON DYNAMICS

E. Del GIUDICE¹, S. DOGLIA² and M. MILANI³*Istituto di Scienze Fisiche dell'Università di Milano, Via Celoria 16, 20133 Milan, Italy*

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We discuss the occurrence of self-focusing of Fröhlich electric vibrations in living cells by investigating the Kerr activity of actin solutions. This process could provide a dynamical support of the filamentous microstructure of cell cytoplasm.

The aim of this letter is to sketch a mechanism which could account for the filamentous microstructures observed in the cytoplasm of eucaryotic cells. We suggest that these structures are dynamically produced when a Fröhlich activity is somehow switched on inside a solution of the proper macromolecules.

Modern cytology has discovered that cell cytoplasm is not an aqueous solution of macromolecules, as it was believed some time ago, but a system of loosely bound networks of microtubules, microfilaments and microtrabeculae immersed in water which fills up the interstitial space [1,2]. This cytoskeleton is not a static structure but changes continuously its shape and size with time, cell activity and also with external stimuli. Its main components are polymers of proteins with high permanent dipole moments, tubulin in the microtubules, and actin in the microfilaments.

An active equilibrium between polymer and monomer forms takes place [3]. Actually a dynamical process called treadmilling has been observed for microtubule polymerization "in vitro", when a continuous disassembly of the monomers at one end and an assembly at the other end of the polymer occurs, thus realizing a directional flux of tubuline along the polymer in the medium [4]. This filament network, that realizes a dynamical connection between membranes, organelles and nucleus in the cell, looks like "a vectorial framework upon which some other force

generating component is organized" [5].

Signals apparently connected with cell dynamics have been detected in metabolically active cells by microwave and laser Raman spectroscopies [6–10]. It has been pointed out that their frequencies are integer multiples (higher harmonic generation), sums (frequency mixing) and differences (parametric generation) of a few fundamentals [7,11,12]. These nonlinear vibrations have been discussed in the literature [6,12–14]; we take the point of view that they are the coherent *electric* vibrations predicted by Fröhlich which are expected to exist at appropriate times of the cell life cycle [13].

The generation of the whole spectrum from a few fundamentals implies, therefore, a nonlinear relationship between the polarization P of the medium and the electric field E of the Fröhlich vibrations, which can reach rather high values (up to 10^7 V/m):

$$P/\epsilon_0 = P(E)/\epsilon_0 = \chi_{ij}E_j + \chi_{ijk}E_jE_k + \chi_{ijkl}E_jE_kE_l + \chi_{ijklm}E_jE_kE_lE_m + \chi_{ijklmn}E_jE_kE_lE_mE_n + \dots \quad (1)$$

In a simplified way (neglecting the vectorial character of E) we may write eq. (1) as

$$P = \epsilon_0 \chi(E)E. \quad (1')$$

The corresponding expansion of the complex dielectric constant in powers of the field is

$$\epsilon = \epsilon_0 + \epsilon_2 |A|^2 + \epsilon_4 |A|^4 + \dots,$$

where

¹ Istituto Nazionale di Fisica Nucleare, Sezione di Milano.

² Gruppo Nazionale Struttura Materia del CNR, Milano.

³ also: Istituto Nazionale di Fisica Nucleare, Sezione di Milano.

$$E = \frac{1}{2} A(r, t) \exp[i(\omega_0 t - kr)] + \text{c.c.}, \quad (2)$$

ϵ_0 is the linear dielectric constant (the asymptotic value of ϵ as $A \rightarrow 0$), $\epsilon_2, \epsilon_4, \epsilon_6, \dots$, are complex quantities whose real parts can have, in general, arbitrary signs.

From the Clausius—Mossotti equation for an isotropic medium [15]

$$(n^2 - 1)/(n^2 + 2) = (1/3\epsilon_0)\chi = \frac{4}{3}\pi\rho\alpha, \quad (3)$$

where χ is the electric susceptibility, ρ is the density of the medium, and α the polarizability of the molecules, it follows that the refractive index n of the medium can be written as

$$n = n_0 + n_2|A|^2 + n_4|A|^4 + \dots, \quad (4)$$

which points out that n depends on the intensity of the propagating wave. This nonlinear dependence can be connected with various physical causes such as changes in the polarizability of the molecules, electrostriction or eventually heating [16]. In liquids with anisotropic molecules the higher powers of E appearing in eq. (4) arise from the alignment of the molecules in the field. The validity of eq. (4) in a given isotropic medium can be tested by means of the Kerr effect, namely by measuring the birefringence induced by an applied electric field E . The difference between the refractive indices of the medium $n_{\parallel}$ parallel and $n_{\perp}$ perpendicular to the direction of the applied electric field is found to be

$$\Delta n \equiv n_{\parallel} - n_{\perp} = K\lambda_0 E^2, \quad (5)$$

where K is the Kerr constant, λ_0 is the vacuum wavelength of the probe beam; the optical phase retardation is

$$\Delta\phi = l(2\pi/\lambda_0) \Delta n, \quad (6)$$

l being the length of the probe beam in the sample. Kerr activity has been observed for aqueous solutions of actin both in the monomer (G-actin) and polymer (F-actin) forms [17,18]. Specifically, positive birefringence has been observed for G-actin and small polymers (of length $L < 1.5 \mu\text{m}$). F-actin (long polymers, $L > 1.5 \mu\text{m}$) exhibits negative birefringence for values of the electric field $E < 5.5 \times 10^4 \text{ V/m}$, whereas higher fields favours the appearance of positive birefringence. In the F-actin case the observed birefringence is the sum of two components of oppo-

site signs which can be distinguished on the basis of their different decay times. The positive component increases with the square of the field while the negative one saturates at high fields; the two balance around $E = 5.5 \times 10^4 \text{ V/m}$ [17].

It follows that for F-actin the regime of eq. (4) with $n_2 > 0$ is reached only after a rearrangement of the dipoles responsible for the negative contribution to the birefringence, which eventually saturates. This peculiar behaviour shows that the occurrence of positive birefringence implies a reorientation of the monomers and a redistribution in the polymer.

In conclusion Kerr measurements hint that for values of the electric field E of the order of magnitude expected to occur in living cells positive birefringence is present both for G- and F-actin.

An electric signal E propagating in an isotropic Kerr active medium induces therefore a change in the refractive index proportional to the intensity of the beam, i.e. $n = n_0 + n_2 E^2$. In a medium with $n_2 > 0$ such as CS_2 , nitrobenzene, bromobenzene, etc. [19] the theory of self-focusing developed in refs. [16,20] shows that an incoming beam shrinks and after a length R_{nl} becomes confined within a thin waveguide whose length R is an increasing function of E . For distances greater than R diffraction effects become predominant again and the beam spreads out. We get from the theory that R_{nl} , the parameter describing the focusing properties of the medium is

$$R_{\text{nl}} = a(\epsilon_0/2\epsilon_2 E_0^2)^{1/2}, \quad (7)$$

a being the initial radius of the beam and E_0 the field intensity on the beam axis.

A rather crude description of the phenomenon in the geometrical-optics approximation points out also the existence of another parameter P_{cr} , that rules the disappearance of diffraction spreading of the beam:

$$P_{\text{cr}} = (1.22)^2 \lambda_0^2 c / 256 n_2. \quad (8)$$

Actually, for large "supercritical" power P ($P \gg P_{\text{cr}}$), when the behaviour of the beam is well described by the geometrical-optics approximation, it can be shown that the wave may be trapped without spreading at any arbitrary diameter.

From the above discussion, the positive Kerr birefringence displayed by actin solutions implies that self-focusing effects and filamentation are also expected to occur when Fröhlich electric vibrations are

produced in the system by a suitable molecular dynamics [19].

Considering now the difference between the index of refraction inside the filament produced by a Fröhlich wave ($n = n_0 + n_2 E^2$) and outside it ($n = n_0$), a positive gradient in the electric susceptibility χ of the medium arises in the direction normal to the filament axis, i.e. to the direction of propagation of the signal.

At this point it is interesting to inquire which consequences on the medium, i.e. on a solution of actin monomers, can be produced by the filamentation of a Fröhlich wave. The volume force acting on a charge distribution near the filament is given by [21]

$$F_v = \rho E - \frac{1}{2} \epsilon_0 E^2 \nabla \chi + \frac{1}{2} \epsilon_0 \nabla [E^2 (d\chi/dg)g], \quad (9)$$

where ρ (g) is the charge (mass) density. (Strictly speaking this formula is valid for static fields; the time dependent expression would only involve a more complex formulation).

The second term in expression (9) represents a force which will appear whenever an inhomogeneous dielectric is placed in an electric field; the last term represents, instead, a volume force acting on a dielectric in an inhomogeneous electric field. In the presence of high fields E , the second and third term will play a dominant role.

For a G-actin solution at high value of E the second term dominates; actually the third term, being a pure gradient, does not contribute to the net volume force but it is responsible for local charge fluctuations. Free actin monomers in the neighbourhood of a filament will therefore experience an attractive force and will consequently align themselves along the filament with their dipole moments oriented transversally to it. Under suitable conditions this alignment can lead to the formation of a polymer structure, F-actin.

In conclusion Fröhlich coherent electric waves, propagating inside a proper molecular solution, can provide a polymerizing mechanism among polar

molecules. This suggests that they could actually be the dynamical agent responsible for the formation and evolution of the cytoskeleton during the cell life cycle.

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